

Functional and Performance Testing

Filled project document for GitHub and SkillWallet submission

Date	28 June 2026	Team ID	To be updated in SkillWallet
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction	Team Members	Devireddy Prasanna, Chokka Abitha, Navya Borugadda, Bodi Varshitha, Mulakala Harika Venkata Sridevi

Functional Test Cases

TC	Feature	Test Scenario	Expected Result	Status
TC-01	Input form	Enter valid weather values.	Prediction page displays Flood or No Flood.	Pass
TC-02	Validation	Submit empty or non-numeric values.	Application asks for valid input.	Pass
TC-03	Model loading	Start Flask app with model.pkl and scaler.pkl.	Artifacts load without errors.	Pass
TC-04	Prediction	Use sample high rainfall values.	System returns high flood risk classification.	Pass
TC-05	GitHub	Open repository after upload.	All documents, code, dataset, and report are visible.	Pending final upload

Performance Testing

Metric	Target	Observed / Expected
Prediction response time	Under 2 seconds for one request	Expected under 1 second after model is loaded.
Model accuracy	Highest-performing model selected	XGBoost scenario accuracy: 96.55%.
Application startup	No missing dependency errors	requirements.txt should include Flask, joblib, pandas, scikit-learn, xgboost.
Usability	Simple input and clear result	Dashboard, form, result page.